

ALI SHAHIDI

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SUMMARY

Software Developer specializing in ASP.NET, C#, Python, React, and MySQL with experience building web and mobile/XR/desktop applications. Skilled in writing clean, efficient code, optimizing application performance, and developing scalable solutions. Eager to learn new technologies and continuously improve development practices.

PROFESSIONAL EXPERIENCE

Software Developer

September 2024 - November 2025

ETHEREAL Research Group

Calgary, AB

- Collaborated in the design of an interactive VR training environment for autistic users, focusing on accessibility, interaction flow, and ethical design.
- Engineered backend systems to synchronize GPT-driven dialogue, user events, and sensory feedback in real time across multimodal inputs.
- Incorporated and implemented AI-driven real-time behavioral analysis, boosting video analysis speed by 90% and accuracy by 83% with local algorithms.
- Optimized performance by reducing third-party API costs by 98% through adaptive frame rate control and delta-based body language analysis.

Software Developer - Protégé

September 2024 - March 2025

ELIXR & Illumia

Calgary, AB, Remote

- Contributed to 5 cross-functional projects, delivering a scalable solution for VR/AR experiences. Focusing on areas such as animation, lip-syncing, AI integration, performance tuning, real-time audio streaming, real-time Speech-to-Text processing, and interactive WebXR, Mobile, VR/AR, and PC experiences.

Software Developer - Internship

February 2024 - August 2024

Telus

Calgary, AB

- Collaborated with Telus Communications to develop a cross-platform AR application for iOS and Meta Quest 2.
- Engineered multiplayer functionality with NetCode and spatial anchor integration for AR app on iOS/Meta Quest 2.

Graduate Research & Teaching Assistant - Software Engineer

January 2021 - April 2024

University of Calgary

Calgary, AB

- Led a study on assistive XR technology in collaboration with psychologists and professionals.
- Provided Unity support during a 2-day IEEE Tech-Hackathon (SIGHT).
- Delivered courses on Software Engineering, OOP, Scalable Data Analysis, and Programming Fundamentals.

EDUCATION

University of Calgary

April 2024

Master of Software Engineering (GPA: 4/4.00)

Calgary, AB

- Received full departmental graduate funding and international graduate tuition award.
- Nominated for the Best Graduate Project at the Distinguished Master's Thesis Award in STEM Disciplines.

University of Tehran

July 2020

B.Sc. in Computer Engineering (GPA: 17.24/20.00)

Tehran, Iran

- Received the best undergraduate project, and two Faculty of Engineering Fellowship Awards.

PROJECTS

Dating App | ASP.NET Core, MVC, REST APIs, MSSQL, Dapper, Entity Framework

- Built a full-stack backend using ASP.NET Core and MSSQL following MVC architecture, implementing REST APIs for user management (GET, POST, PUT, DELETE) with Swagger documentation.
- Implemented secure JWT authentication with password hashing/salting, 24-hour token expiration, and role-based access for users and admin tools.
- Designed the data access layer using Dapper and Entity Framework, enabling scalable CRUD operations, efficient SQL Server queries, and support for likes/matches, messaging, and photo management.

ETHEREAL Research Group - Coffee Shop Simulation | Unity, C#, .NET, Chat GPT, API, VR, Eleven Lab

- Built an immersive VR coffee shop simulator to help autistic users practice communication and decision-making in a realistic social setting.
- Implemented backend services using ASP.NET and MySQL to support dynamic application behavior and API endpoints for data management and communication.
- Integrated GPT, ElevenLabs, and CC5 avatar creation for real-time avatar narration, adaptive dialogue, and event-driven responses within the virtual space.
- Implemented GPT-based image-to-text parsing to interpret menu visuals and generate a binary decision tree UI for accessible, guided item selection and checkout.

ETHEREAL Research Group - Multi Modal Behaviour Analysis | Unity, AI, Azure, VR/AR/Tablet, Python

- Built a cross-platform real-time video analysis app for tablets and Quest 3, enabling multimodal (text, video, voice, image) input processing.
- Designed a server-based AI pipeline to offload VR computation and support large-file chunking for efficient, reliable data transmission.

ELIXR & Illumia | Unity, C#, Network, Optimization, VR, Python, RAG, API

- Implemented Retrieval-Augmented Generation (RAG) pipelines integrating Llama 3 / Ollama to enable AI models to access domain-specific PDFs and manuals, enhancing knowledge retrieval and automated insights for end-users.
- Built real-time audio system with always-on mic and event-driven responses.
- Built AI-driven animation system using third-party APIs to reflect user emotions.
- Improved performance via Profiler, GPU Instancing, Static Batching, Occlusion Culling, and Object Pooling.
- Real-time data streaming enabled with WebSocket integration in Unity.
- Refactored and scaled codebases for projects including flight simulators, training tools, and virtual assistants.

Telus | Unity, C, Python, Azure

- Led the development of a multiplayer Augmented Reality application with Telus Communications, featuring NetCode, mapping, mesh rendering, animation, and interactions.
- This project will reduce transportation costs and enable remote instruction, enhancing efficiency and user privacy.

SKILLS

Languages: C#, Python, C, C++, Unity, JavaScript

Frameworks: ASP.NET, Express.js, React.js

Databases: MS SQL, MySQL, PostgreSQL

Operating Systems: Linux, Windows

Tools: Git, ClickUp, Docker, ARKit, ChatGPT, Cursor, Scrum, Jira, Confluence, LaTeX

CERTIFICATIONS

C# .NET with MS SQL Complete Beginner to Master 2026 | Udemy

March 2026

Programming Design Patterns For Unity: Write Better Code | Udemy

July 2024

The Beginner's Guide to Animation in Unity | Udemy

June 2024

The Complete Web Development Bootcamp | Udemy

January 2024